SUPPLY CHAIN ANALYTICS
DATA-DRIVEN BUSINESS OPTIMIZATION

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Project Timeline: Nov’25 to Dec’25

DATASET OVERVIEW

ROWS: 500
COLUMNS: 20

Categorical Columns

- Warehouse_ID
- Location
- Supplier_ID
- Product_Category

Numerical Columns

- Current_Stock
- Demand_Forecast
- Lead_Time_Days
- Shipping_Time_Days
- Stockout_Risk
- Operational_Cost
- Monthly_Sales
- Order_Processing_Time
- Return_Rate
- Customer_Rating
- Warehouse_Capacity
- Storage_Cost
- Transportation_Cost
- Backorder_Quantity
- Damaged_Goods
- Employee_Count
- Stock_Difference
- Forecast_Accuracy_Percent
- Profit / Loss

TOOLS & TECHNIQUES USED

- Python – Data cleaning, validation, and exploratory analysis
- SQL (MySQL) – Objective-wise data analysis and creation of analytical views
- Power BI – Interactive dashboards and business insights visualization

BUSINESS CONTEXT

The company is facing end-to-end supply chain inefficiencies:

- Delayed deliveries
- Inventory imbalance
- Poor demand forecasting
- High operational cost & losses

OBJECTIVE 1: MINIMIZE DELIVERY DELAYS

Insights

- Avg Shipping Time ≈ 3.95 days
- Max Shipping Time = 7 days
- 25% of warehouses are taking 7 days to deliver orders
- Locations like Denver, New York, Miami, Chicago, San Francisco have higher average

Solution

- Optimize delivery routes or use faster transport modes
- Focus on slow warehouses first
- Improve logistics planning region-wise
- Warehouses which have max shipping days: WH008, WH013, WH040, WH042, WH063, WH083, WH134, WH271, WH295, WH298, WH313, WH320, WH350, WH351, WH352, WH358, WH362, WH382, WH413, WH414, WH417, WH444, WH449, WH453, WH470, WH472

OBJECTIVE 2: ENHANCE INVENTORY MANAGEMENT

Insights

- Overstock warehouses: 218
- Understock warehouses: 259
- 25% warehouses have more than 224 backorders
- Half of the products are short by 209 units or more.
- Miami and Atlanta are the most understocked locations

Solution

- Redistribute excess stock from overstocked to understocked warehouses
- Prioritize inventory replenishment for Miami and Atlanta
- Use backorder quantity as an early indicator for stock rebalancing
- Align inventory planning more closely with demand forecasts
- Reduce holding costs by avoiding unnecessary overstock

OBJECTIVE 3: IMPROVE DEMAND FORECASTING

Insights

- Average Demand Forecast = 2957 units, with very high variation (±1399 units), indicating inconsistent forecasting
- Backorders persist even where demand is forecasted to be high. WH449, WH134, WH453, and WH483 have the maximum backorders.
- Dallas, Los Angeles, Denver, and Miami show the highest average backorders (≈160–180 units)
- Denver- Groceries combination shows the highest backorders (~205 units)
- Warehouses with higher Stockout Risk (15–25) have ~150–175 units average backorders, confirming that higher stockout risk is strongly associated with higher backorders

Solution

- Apply location and product-level demand forecasting instead of global averages
- Use backorder quantity and stockout risk as feedback inputs to adjust forecasts
- Prioritize forecast model improvement for Dallas, Los Angeles, Denver, and Miami
- Continuously validate forecasts against actual demand and backorders

OBJECTIVE 4: MAXIMIZE PROFIT MARGINS

Insights

- Company is operating at a net loss of −39.41M
  (Total Cost 42.08M vs Total Sales 2.67M)
- Operational cost: 65886.354
- Transportation Cost: 8001.086
- Storage Cost: 10263.834
- Highest loss locations (avg loss per warehouse): Los Angeles, Miami, Seattle
- Lowest loss locations (comparatively better): Chicago, Dallas
- Product-wise profitability (avg loss): Apparel: −82.2K (worst), Automotive:-78.7 K, Electronics: −77.8K (high sales but still loss-making)
- Critical loss-making product–location pairs:
  - Electronics – Houston: −93.2K
  - Electronics – Los Angeles: −91.2K
  - Apparel – Los Angeles: −90.8K
- High-revenue products (e.g. Electronics ~5600 avg sales) still show losses → profit issue is cost-driven, not revenue-driven

Solution

- Reduce Operational Costs first
- Control Storage Costs by reducing overstock
- Optimize Transportation Routes & Vendors
- Focus cost optimization on high-loss locations
  → Los Angeles, Miami, Seattle show the highest average losses
- Re-evaluate pricing & sourcing for loss-heavy products
  → Apparel and Automotive are the most loss-making categories
- Track profit at product–location level instead of total sales

OVERALL BUSINESS IMPACT

Value Delivered

- Root causes identified (not just symptoms)
- Data-driven solutions proposed
- Decision-ready dashboards created

CONCLUSION

- Delivery delays, inventory imbalance, inaccurate demand forecasting, and high costs were identified as key supply chain issues.
- Data analysis showed that misallocated inventory, forecasting errors, and cost inefficiencies are the main drivers of poor performance and losses.
- Using Python, SQL, and Power BI, actionable insights were generated to improve delivery efficiency, balance inventory, refine forecasting, and optimize profitability.

THANK YOU